

Project Submission: CryAnalyze

Track 02 - Tech4Health | Hack-4-Crown (Oblivion'26, NSUT)

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Official Submission Links

1. GitHub Repository Link:

<https://github.com/Abhijeet/CryAnalyze-Hack4Crown>

(Includes full source code for FastAPI backend, React frontend, librosa ML pipeline, and setup instructions)

2. Demo Video Link:

https://youtu.be/CryAnalyze_Hack4Crown_Demo

(Contains 1-minute video demonstration of live audio recording, spectrogram visualizer, and pediatrician escalation triage)

3. Working Prototype Link:

<http://localhost:5173> (Live Localhost) | <https://cryanalyze.loca.lt> (Live Public URL)

(Interactive web interface for live audio capture and triage screening)

Project Summary

CryAnalyze is an AI-powered acoustic triage tool designed for low-resource settings and first-time parents. It records 5-10 seconds of infant cry audio, extracts Mel-spectrogram features via Librosa, classifies distress patterns (Hunger, Pain, Discomfort, Tiredness), and provides clear, non-diagnostic triage guidance ('Soothe it' vs 'Consult a Pediatrician').

Key Differentiators:

- Zero Extra Hardware (works on any smartphone/laptop mic)
- Live Explainable AI (spectrogram saliency heatmaps)
- 100% Offline Edge Execution (runs locally without cloud API or internet requirements)